

Depression of freezing point of the solvent

1. (c) $\Delta T_f = K_f \times \text{molality} = 1.86 \times 0.05 = 0.093^\circ\text{C}$

Thus freezing point = $0 - 0.093 = -0.093^\circ\text{C}$.

2. (c) $\Delta T_f = \frac{100 \times K \times w}{m \times W} \therefore 0.186 = \frac{100 \times 18.6 \times w}{60 \times 500}$

$$w = 3g$$

3. (a) Camphor has the maximum value of $K_f (= 39.7)$.

4. (d) The extent of depression in freezing point varies with the number of solute particles for a fixed solvent only and it is a characteristics feature of the nature of solvent also. So for two different solvents the extent of depression may vary even if number of solute particles be dissolved.

5. (c) **Decreases the freezing point of water**

Explanation:

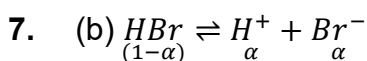
When a non-volatile solute is dissolved in water:

- vapour pressure decreases
- boiling point increases
- freezing point decreases

Therefore, only option (c) is correct.

6. (a) Molar mass = $\frac{K_f \times 1000 \times w}{\Delta T_f \times W} = \frac{1.86 \times 1000 \times 1.25}{20 \times 1.1}$

$$= 105.68 = 105.7.$$



$$\text{Total} = 1 + \alpha \therefore i = 1 + \alpha = 1 + 0.9 = 1.9$$



$$\Delta T_f = iK_f \times m = 1.9 \times 1.86 \times \frac{8.1}{81} \times \frac{1000}{100} = 3.53^\circ C$$

$$T_f = -3.53^\circ C.$$

8. (b) $\Delta T_f = K_f \times m = 1.86 \times 0.1 = 0.186.$

9 (b) Freezing point is lowered on addition of solute in it..

10. (d) $\Delta T_b = 0.18$; $\Delta T_b = mK_b$

$$\frac{0.18}{\Delta T_f} = \frac{mK_b}{mK_f}, \frac{0.18 \times 1.86}{0.512} = \Delta T_f; \Delta T_f = 0.653$$

$$T^0 - T_s = 0.653; T^0 - T_s = 0.653; T_s = 0 - 0.653^\circ C.$$

11. (a) $\Delta T_f = 1.86 \times \left(\frac{342}{342}\right) = 1.86^\circ$; $\therefore T_f = -1.86^\circ C.$

12. (b) $\Delta T_b = K_b \times m$ i.e. $0.52 = 0.52 \times m$

$$\Delta T_f = K_f \times m = 1.86 \times 1 = 1.86; T_f = -1.86^\circ C.$$

13. (b) For NaCl $i = 2$

$$\Delta T_f = 2K_f m = 2 \times 1.86 \times 1 = 3.72$$

$$T_s = T - \Delta T_f = 0 - 3.72 = -3.72^\circ C$$

14. (b) $3.8^\circ C$

Explanation:

Heavy water (D_2O) has a higher freezing point than ordinary water because deuterium forms slightly stronger bonds than hydrogen.

15. (b) $\Delta T_f = K_f \times m \Rightarrow 0.186 = 1.86 \times m$

So $m = 0.1$, Put the value of m in $\Delta T_b = K_b \times m$

$$\Delta T_b = 0.521 \times (0.1) = 0.0521$$



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Explanation:

For dilute solutions:

$$\Delta T_f = K_f m$$

Therefore,

$$\frac{\Delta T_f}{m} = K_f$$

Since K_f is neither 0, 1, nor 3 generally,

(d) None of the above

17 (a) Below 0°C **Explanation:**

A solute lowers the freezing point of water.

18 (c) Boiling point increases and freezing point decreases

Explanation:

Sugar is a non-volatile solute:

- boiling point increases
- freezing point decreases

19 (a) Liquid solvent and solid solvent

Explanation:

At freezing point, solid solvent and liquid solvent coexist in equilibrium.

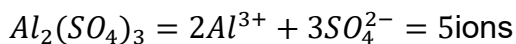
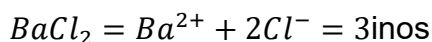
$$20. \text{ (a) By using, } m = \frac{K_f \times 1000 \times w}{\Delta T_f \times W_{\text{Solvent}}(\text{gm})} = \frac{5.12 \times 1000 \times 1}{0.40 \times 50} = 256 \text{ gm/mol}$$

Hence, molecular mass of the solute = 256 gmmol^{-1}

$$21. \text{ (a) } m = \frac{K_f \times w \times 1000}{\Delta T_f \times W} = \frac{5.12 \times 0.440 \times 1000}{0.567 \times 22.2} = 178.9$$

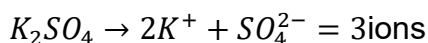


22. (a) $KBr = K^+ + Br^- = 2 \text{ ions}$



$\therefore$ urea is not ionise hence it is shows highest freezing point.

23. (d) $NaCl \rightarrow Na^+ + Cl^- = 2 \text{ ions}$



K_2SO_4 give maximum ion in solution so it shows maximum depression in freezing point.

24. (c) $\Delta T_f = \frac{K_f \times 1000 \times w}{m \times W} = -0.216^\circ C$

25. (a) $\Delta T_f = \frac{1000 \times 1.86 \times 17}{46 \times 1000} = 0.69^\circ C$

$$T_f = 0 - 0.69 = -0.69^\circ C$$

26. (ad) The depression of freezing point is less than that of pure solvent and only solvent molecules solidify at the freezing point.

27. (b) $K_f = \frac{RT_f^2}{1000 \times L_f}, R = 8.314 JK^{-1} mol^{-1}$

$$T_f = 273 + 16.6 = 289.6 K ; L_f = 180.75 Jg^{-1}$$

$$K_f = \frac{8.314 \times 289.6 \times 289.6}{1000 \times 180.75}$$

